Final Project: Pick a Project

Computer Science - Week 10

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Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both) but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Emma Vernon	ejvernon@udel.edu
Friend (optional)		

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

The specific goal I'd like to address with this project is to create a site that allows crocheters to input a working pattern one stitch at a time as they develop a new pattern so that at the end of their brainstorming the site will generate the completed pattern with every detail for them to reuse.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

- *My Row Counter App:*   - *https://rowcounterapp.com*   - *This app will keep a running count of what row you are on (part of what I would like my website to do)* - *Pocket Crochet:*   - *https://pocketcrochet.app/*   - *Not exactly the same but keeps patterns in one place (essentially database for patterns)* - *Crochet.sites:*   - *https://crochets.site/*   - *User inputs what they want to create, and the AI tool will generate a pattern for*

them (not exactly the same thing as I want to do with my website design but is similar)

- *There are multiple crochet sites that will do the row counting, but that is for making the pattern not designing. I couldn't find any websites that didn't take an already pre-existing pattern to start tracking progress.*

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

Crochet patterns take a lot of trial and error to create. Most crocheters do this process on paper. They'll have to write and erase constantly until their pattern is finished. The goal of this website would be to minimize that process so that all they must do is click a button to track their pattern progress. It would be useful so that they wouldn't have to worry about losing the paper, ripping the paper, running out of room, spilling something on the paper, etc.

I want the web application to be used so that they can create their initial working pattern by inputting the details (i.e. the number of rounds, types of stitches, stitch count, etc.), add what kind of hook and hook size, what yarn type, color and the amount they used, if they added additional features (felt, safety eyes/noses, buttons etc.) and where, and then as they work along they can make edits to the rows that they had already completed as the tweak their patterns. At the end of the brainstorming, the site will print out all the instructions in an easy-to-follow format

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The core goal is to make the process of designing a crochet pattern simpler and more efficient.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

- Row count
- Row instructions

- Total Stitch count
- Type of stitch
- Yarn
    - Color
    - Type
    - Amount
    - cost?
- Hook
    - Size
    - Type?
- Details
    - Are there color changes, safety eyes, felt, etc.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

The computer will manipulate the data the user inputs by:
- Calculating the stitch breakdown when given total stitches that you end with and row instruction (i.e. row ends with 12 stitches, increasing around = program writes 6 increase stitches)
- Will store what kind of yarn, details and hook the user used and add to instructions
- Count how many total rows there are
- Loop through each of the rows to generate the final instructions
- Takes the user input for what kind of stitch, number of stitches, row, and any special instructions

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The user will get back an easy to follow, completed pattern

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What is the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

Nouns the program needs to track:
- *Row*
- *Stitch*
- *Details*
- *Yarn*
- *Hook*

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

- *Row*
    - *Row count*
    - *Row instruction*
- *Stitch*
    - *Stitch count*
    - *Stitch type*
- *Details*
    - *Type*
- *Yarn*
    - *Type*
    - *Amount*
    - *Cost - abstract away*
    - *Color*
- *Hook*
    - *Type - abstract away*
    - *Size*

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

The most difficult part of the project will be organizing how the code should look and run and trying to cover all bases. It's going to be a longer project, and without exact instructions it's going to be very open ended. I need to make sure the program I create is in a logical order and that I have everything that the rubric requires without use of an assignment that guides me through it.

The most technically complex part might be designing the routes so that everything lines up with what it needs to do. The routes are in my opinion the most crucial part of creating a website, so if I mess up there it would throw my entire project off. I think it will take me the most time between coming up with what routes to use to coding it.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I will likely be using an LLM to help me code. To avoid shallow learning, if the LLM gives me a piece of code, rather than plugging that in I'll ask to follow up questions to ensure that I understand what it put out. From there I'll try to code my own version of what the LLM produced using what it initially put out as well as my follow up questions and answers. If I am struggling with a concept or section of code, I'll have it produce practice problems that I can use to build that skill up.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

I would like to use this for myself and my friends. We are all avid crocheters. I have not made a pattern myself, but one of my friends has expressed to me a little while ago that making patterns is super tedious and she went through multiple sheets of paper just trying to get the pattern to look as she envisioned it. I hope I can possibly share this with her if it works well so that it would make her design process easier. I also hope this might encourage me to start making my own patterns because I think that the process is a bit much to do and it was easier just to look up pre-made patterns even if it wasn't exactly what I was looking for.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours.*

What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"). Paste your prompt below:

Asked LLM:

I want to build a simple website that will make planning/designing a crochet pattern easier by letting the user input what row they are on, what stitch they are using, how many total stitches, are they changing colors, and other related needs so that the website will generate a final pattern based on their inputs. What are 3 three simple features a basic website should have and what are 3 common problems crocheters face when they are in the design process

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

LLM Response Summarized:

- *Three simple features:*
  - *Custom pattern generator*
    - *User inputs various data representing their instructions and that gets converted into a readable pattern (how many rows, how many stitches, etc.)*
  - *Row and stitch tracker*
    - *User records current row/stitch count (using buttons)*
  - *Color change and pattern notes*
    - *User inputs details relating to the pattern (ex: changing colors) that gets added to the generated pattern*

- *Three common problems crocheters face during design process:*
  - *Losing track of row number and stitch counts*
    - *Website will have a built-in tracker*
  - *Turning design ideas into clear written instructions*
    - *Website solution: pattern will be generated as an easy-to-read format*
  - *Managing color changes and stitch adjustments*
    - *Website solution: able to add to the pattern when these changes occur*

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

Asked LLM:

The color change and pattern notes may be too complex for an introductory programming project. There are a lot of details that go into a pattern, not just a color change, so it would be difficult to make cases for everyone so that a whole line of instructions are created from a couple of short inputs. Is there a simpler alternative that would have the same goal?

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

My project idea has changed slightly. The LLM recommended using predefined options that the user can choose from to generate the pattern. Initially I planned on letting them input certain conditions and then the code will take that information and store it where it needs to be. In the LLM's example, it would look almost like a google form, where there would be text inputs for row amount and number of stitches and then check boxes for what kinds of stitches and if decreasing or increasing stitches. I'll have to continue to revise this idea as it still does not quite address the issue of more detailed instructions. But for a simple pattern, this looks like it will generate a pattern for the user with less complexities for actual coding.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

I plan to utilize office hours and discord to reach out to TA if I find I am having trouble with the project. I'll also look through the blockpy textbook and drafter documentation to see if I can find examples or anything matching what I could be stuck on. If I am stuck on a concept or need further explanation of what a function does or what I need it to do, if allowed I'll consult the LLM provided as well.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

To try and address an issue that needs debugging, I would go to thonny and identify where there might be an error in the code. I'll take what thonny will produce and try to address the issue using resources such as the blockpy textbook, drafter documentation, consult the LLM once more, or if I cannot solve the issue on my own reach out to a TA during office hours or on Discord to receive further guidance.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I want to build a simple website that will make planning/designing a crochet pattern easier by letting the user input what row they are on, what stitch they are using, how many total stitches, are they changing colors, and other related needs so that the website will generate a final pattern based on their inputs. What are 3 three simple features a basic website should have and what are 3 common problems crocheters face when they are in the design process

For your crochet pattern generator website, I would focus on features that help crocheters organize their pattern, keep track of their progress, and avoid mistakes while designing. Since this is a CS1 project using Drafter, you can build a useful version with simple inputs, calculations, and generated text without needing a database or backend.

1. Three simple features your website should have
Criador de Padrões de Crochê com IA: Crie, Visualize e Baixe

1. Custom Pattern Generator

Let the user enter the row number, stitch type, total stitch count, and instructions such as increases or decreases. The website combines those inputs into readable crochet pattern instructions.

Example output: “Row 3: Single crochet in each stitch around (24 stitches).”

Strickkurse – KNIT KNIT Berlin

2. Row and Stitch Tracker

Allow users to record their current row and stitch count, then update their progress as they work. Include a simple next-row button or editable row number.

This helps crocheters remember where they stopped and keep track of how many stitches they should have.

Granny Square Color Change Tip: Reduce Yarn Tails by Half! | Creative Crochet Corner

3. Color Change and Pattern Notes

Let users specify whether they are changing colors, choose a color name, and add notes about where the change happens. Include those details in the generated pattern.

Example output: “Row 5: Switch to blue yarn at the beginning of the row.”

2. Three common problems crocheters face during the design process
1. Losing track of row numbers and stitch counts

Crocheters can lose their place, forget how many stitches they should have, or accidentally add or skip stitches. This can cause a project to become uneven or turn out differently than intended.

Your website's solution: A row tracker and a field for the expected stitch count.

2. Turning design ideas into clear written instructions

When creating an original project, crocheters may know what they want to make but struggle to write instructions that are organized, consistent, and easy to follow.

Your website's solution: A pattern generator that turns form inputs into standardized row-by-row instructions.

3. Managing color changes and stitch adjustments

Changing colors, increasing or decreasing stitches, and keeping track of where these changes occur can make a pattern harder to organize and follow.

Your website's solution: Options for color changes, increases, and decreases that are included in the generated instructions.

3. How to keep the project realistic for CS1
I'd recommend building the website around one simple workflow:

1. User inputs

Row number, stitch type, stitch count, color change, and adjustments

2. Program processes inputs

Checks the inputs and combines them into pattern instructions

3. Generated crochet pattern

Readable row-by-row instructions the user can copy or save

For a first version, use Drafter forms, conditional statements, string formatting, and lists to assemble the pattern. You could also use browser local storage to preserve a draft between visits.

One important limitation: Your website can generate instructions from the user's choices, but automatically guaranteeing that every combination produces a valid crochet pattern is much harder. Start with a few supported stitch types and simple operations, and validate stitch counts where possible.

That gives you a manageable project with a clear input-process-output model, while still solving a real problem for crocheters.

The color change and pattern notes may be too complex for an introductory programming project. There are a lot of details that go into a pattern, not just a color change, so it would be difficult to make cases for everyone so that a whole line of instructions are created from a couple of short inputs. Is there a simpler alternative that would have the same goal?
Yes! A simpler alternative would be to replace Color Change and Pattern Notes with a Pattern Row Builder. Instead of trying to account for every possible crochet instruction, the user would build their pattern one row at a time using a few predefined options.

Recommended alternative: Pattern Row Builder
Example of how it could work
A sample interface for your website

Row number
3
Stitch type

Single crochet

Half double crochet

Double crochet
Total stitches
24
Row instruction

Keep the same stitch count

Increase stitches

Decrease stitches
Generated row
Row 3: single crochet — work until you have 24 stitches.

Target stitch count: 24

Interactive mock-up: demonstrates how a few inputs can produce a basic row description. The generated text does not guarantee mathematically correct crochet instructions.

Why this is simpler
Limited options: You only support a few stitch types and row actions.

Less complicated logic: Use if/elif statements to select the right instruction.

Consistent output: Every row follows the same format.

Easy to expand: You can add more stitch types or actions later if time allows.

Two other simple alternatives
Amimono Notebook(編み物用ノート)(送料無料) | crochet and me

1. Row-by-row pattern organizer

Let users type their own instructions for each row and organize them into one neatly formatted pattern. This avoids needing to predict every possible crochet instruction.

Pembe Tavşan Kostümlü Amigurumi Oyuncak Yapım Şeması | HandmadeV

2. Stitch Count Calculator

Let users enter a starting stitch count and the number of increases or decreases, then calculate the resulting total. This focuses on one specific design challenge rather than generating an entire pattern.

My recommendation
For a three-week introductory programming project, I'd choose the Pattern Row Builder if you want to keep the automatic generation idea, or the Row-by-row Pattern Organizer if you want the simplest and most flexible solution.

The organizer is especially practical because crocheters remain responsible for the actual instructions, while your website handles the tedious work of formatting and organizing them. That still addresses the problem of turning design ideas into a clear, readable pattern without requiring you to program rules for every possible stitch combination.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.